Probability and Statistics

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Outline I

Course Structure

Textbooks

1. *Prob. & Stats. for Engineers and Scientists*, S. Ross (4th Ed.)
2. *Introductory Statistics*, S. Ross

Probability Theory

Building the mathematical framework for uncertainty:

- **Fundamentals:** Axioms, Conditional Prob., Bayes' Theorem, Independence.
- **Variables:** Joint/Single RVs, Expectation, Covariance.
- **Asymptotics:** Markov/Chebyshev Inequalities, Law of Large Numbers.
- **Asymptotic Limit: Central Limit Theorem (CLT)**, Special Distributions (Normal, Binomial, Poisson, t, χ^2).

Statistical Inference

Statistical Inference

Extracting meaningful, data-driven decisions from sample data:

- **Estimation:** Point/Interval estimators, MLE, Confidence Intervals.
- **Hypothesis Testing:** One/Two-sided tests, Significance levels, p-values, ANOVA, Goodness of Fit.
- **Modeling:** Linear Regression, Least Squares, Residual Analysis, Coefficient of Determination (R^2).

Core Learning Outcomes

By the end of this course, you will be equipped to:

1. Map real-world phenomena to discrete and continuous distributions.
2. Design, run, and interpret rigorous statistical hypothesis tests.
3. Apply least-squares and maximum likelihood estimation to engineering datasets.

What is an Experiment?

Definition (Random Experiment)

A **random experiment** (or statistical experiment) is a process, trial, or observation that leads to one of several possible outcomes, where the exact result cannot be predicted with absolute certainty beforehand.

Three Key Characteristics of a Random Experiment

1. **Repeatability:** The process can, at least in theory, be repeated indefinitely under identical conditions.
2. **Unpredictability:** While the conditions are controlled, the outcome of any single trial is governed by chance.
3. **Predefined Outcome Space:** Although we do not know *which* specific outcome will occur, the set of all possible outcomes is fully known before the trial begins.

What is an Experiment?

Examples

- Tossing a coin, rolling a six-sided die, or drawing a card.
- Measuring the round-trip latency ($t \geq 0$ ms) of a packet sent over a network.
- Querying a database server to see if its state is *Active*, *Idle*, or *Overloaded*.

Defining the Outcome Space: The Sample Space

Definition (Sample Space)

The **sample space** of a random experiment, denoted by $\mathcal{S}$ (or the Greek letter Ω), is the set of all possible outcomes of that experiment. An individual result in this set is called an **outcome** or a **sample point**.

Classification of Sample Spaces

Depending on the nature of the experimental outcomes, $\mathcal{S}$ can be classified as:

- **Discrete (Finite or Countable):** Outcomes can be listed in a sequence.
- **Continuous (Uncountable):** Outcomes form an interval of real numbers, representing measurements.

Examples

Examples: Discrete vs. Continuous

- **Discrete:** Tossing a fair coin twice:

$$\mathcal{S} = \{HH, HT, TH, TT\}$$

- **Discrete:** Tracking the number of active processes on a server with limit N:

$$\mathcal{S} = \{0, 1, 2, \dots, N\}$$

- **Continuous :** Measuring the lifetime X (in hours) of a computer chip:

$$\mathcal{S} = \{x \in \mathbb{R} \mid x \geq 0\}$$

Defining Events: Subsets of the Sample Space

Definition (Event)

An **event** is any collection of outcomes of a random experiment. Mathematically, an event E is a **subset** of the sample space $\mathcal{S}$ (denoted by $E \subseteq \mathcal{S}$).

An event E is said to **occur** if the actual observed outcome of the experiment is a member of the set E.

Special Cases of Events

- **Simple (Elementary) Event:** An event containing exactly one outcome (e.g., $E = \{e\}$ where $e \in \mathcal{S}$).
- **Sure (Certain) Event:** The entire sample space $\mathcal{S}$ itself. It is guaranteed to occur on every trial.
- **Null (Impossible) Event:** The empty set $\emptyset$. It contains no outcomes and can never occur.

Examples

Examples

- **Rolling a Fair Six-Sided Die:** $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$
  - Event A (Rolling an even number): $A = \{2, 4, 6\}$
  - Event B (Rolling a number greater than 4): $B = \{5, 6\}$
- **Tossing a Coin Twice:** $\mathcal{S} = \{HH, HT, TH, TT\}$
  - Event C (Getting exactly one head): $C = \{HT, TH\}$
  - Event D (Getting the same result on both tosses): $D = \{HH, TT\}$
- **Lifetime of a Chip (X hours):** $\mathcal{S} = \{x \in \mathbb{R} \mid x \geq 0\}$
  - Event E (The chip lasts more than 1000 hours): $E = \{x \in \mathcal{S} \mid x > 1000\}$
  - Event F (The chip fails within the first 100 hours): $F = \{x \in \mathcal{S} \mid 0 \leq x \leq 100\}$

Set Operations on Events

Definition (Basic Operations)

Let A and B be events in a sample space $\mathcal{S}$. We define:

- **Union ($A \cup B$):** The event that occurs if A *or* B (or both) occur.
- **Intersection ($A \cap B$):** The event that occurs if both A *and* B occur.
- **Complement (A^c or A'):** The event that occurs if A does *not* occur.

Mutually Exclusive (Disjoint) Events

Two events A and B are said to be **mutually exclusive** or **disjoint** if they cannot occur at the same time. Mathematically, this is expressed as:

$$A \cap B = \emptyset$$

Examples

Examples

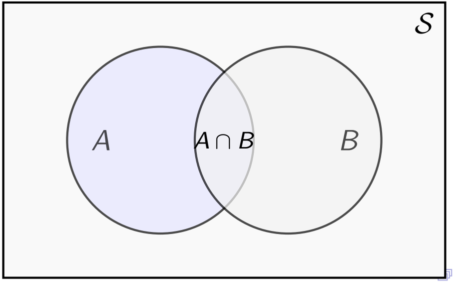
- **Rolling a Die:** Let $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$. If $A = \{2, 4, 6\}$ (even) and $B = \{1, 3, 5\}$ (odd):
  - $A \cap B = \emptyset$, meaning the events are mutually exclusive.
- **System Status Monitor:** Suppose a system is monitored and its state space is $\mathcal{S} = \{\text{Active}, \text{Idle}, \text{Overloaded}\}$. If $E_1 = \{\text{Active}\}$ and $E_2 = \{\text{Active}, \text{Overloaded}\}$:
  - $E_1 \cup E_2 = \{\text{Active}, \text{Overloaded}\}$
  - $E_1 \cap E_2 = \{\text{Active}\}$

Visualizing Events: Venn Diagrams

Venn Diagrams

Venn Diagrams visually represent the relationship between events in a sample space $\mathcal{S}$:

- **Sample Space ($\mathcal{S}$):** Represented by the outer bounding rectangle.
- **Events (A, B):** Represented by closed regions (usually circles or ellipses).



De Morgan's Law

These laws establish fundamental mathematical dualities between unions, intersections, and complements.

1. Complement of a Union:

$$(A \cup B)^c = A^c \cap B^c$$

- *Intuition:* The event that "neither A nor B occurs" is identical to "A does not occur" **and** "B does not occur".

2. Complement of an Intersection:

$$(A \cap B)^c = A^c \cup B^c$$

- *Intuition:* The event that "A and B do not both occur" is equivalent to "A does not occur" **or** "B does not occur".

Exercise 1: Cardinality & Two Fair Dice

Problem Statement

Two fair, six-sided dice are rolled. The sample space contains $|\mathcal{S}| = 36$ equally likely outcomes. Let:

- Event A: The sum of the two dice is exactly 7.
- Event B: At least one of the dice shows a 4.

Find:
(a) $|A|$, (b) $|B|$, (c) $|A \cap B|$, and (d) $|A \cup B|$.

Exercise 2: Card Deck & De Morgan's Laws

Problem Statement

A single card is drawn from a standard, well-shuffled 52-card deck ($|\mathcal{S}| = 52$ equally likely outcomes). Let:

- Event F: The drawn card is a Face Card (Jack, Queen, or King).
- Event H: The drawn card is a Heart ($\heartsuit$).

Find: (a) $|F \cup H|$, and (b) the number of ways that the card is **neither** a face card **nor** a heart using De Morgan's Laws and complement cardinality.

Exercise 3: Three Coin Tosses

Problem Statement

A fair coin is tossed three times.

1. Write out the complete, equally likely sample space $\mathcal{S}$.
2. Let Event A be "obtaining at least two heads" and Event B be "obtaining exactly one tail". Find $|A|$, $|B|$, and $|A \cap B|$.

What is Probability?

Definition (Probability)

Historically and mathematically, **probability** is a numerical measure (ranging from 0 to 1) assigned to an event that quantifies the likelihood of its occurrence.

- A probability of 0 indicates the event is **impossible**.
- A probability of 1 indicates the event is **certain**.

Paradigm 1: The Classical Definition (Equally Likely)

If a random experiment has a finite sample space $\mathcal{S}$ consisting of N outcomes, and we assume all outcomes are **equally likely** to occur:

$$\text{Probability of Event } A = P(A) = \frac{\text{Number of outcomes in } A}{\text{Total number of outcomes in } \mathcal{S}} = \frac{|A|}{|\mathcal{S}|}$$

Quantifying Probability: Empirical & Subjective

Paradigm 2: The Empirical Definition (Relative Frequency)

When outcomes are not equally likely, we estimate probabilities by conducting a random experiment repeatedly. If the experiment is performed n times, and event B occurs n_B times:

$$P(B) \approx \frac{n_B}{n}$$

As the number of trials grows infinitely large ($n \to \infty$), this relative frequency stabilizes to the true probability (formalized later by the *Law of Large Numbers*).

Examples

Classical Example

Consider rolling a fair, six-sided die. The sample space is $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$, so $|\mathcal{S}| = 6$. Let A be the event of rolling an even number, $A = \{2, 4, 6\}$, so $|A| = 3$.

$$P(A) = \frac{3}{6} = 0.5$$

Empirical Example

In testing a manufacturing assembly line, we observe that out of $1,000$ manufactured items, 15 are defective.

$$\text{Estimated Probability of a Defect } = \frac{15}{1000} = 0.015 \text{ (or } 1.5\%\text{)}$$

Comparing Approaches: Equally Likely vs. General

Comparison between two definitions

The primary difference between these paradigms is the assumption of symmetry.

Feature	**Classical**	**Empirical/General**
Assumption	Uniform (All outcomes equal)	Non-uniform (Weighted)
Condition	Symmetry exists	No symmetry required
Calculation	Simple Ratio: $\lvert A\rvert/\lvert\mathcal{S}\rvert$	Frequency: n_A/n
Applicability	Idealized experiments	Real-world data

Despite these differences both of these approaches share some fundamental properties.

Property 1: Boundedness ($0 \le P(A) \le 1$)

The Property

The probability of any event A is restricted to a value between 0 and 1.

Classical Derivation (Set Cardinality)

Since $A \subseteq \mathcal{S}$, the size of A cannot be negative or exceed the total size of $\mathcal{S}$:

$$0 \le |A| \le |\mathcal{S}| \implies 0 \le \frac{|A|}{|\mathcal{S}|} \le 1 \implies 0 \le P(A) \le 1$$

Empirical Derivation (Trial Counting)

The count of occurrences n_A in n trials must lie between 0 and n:

$$0 \le n_A \le n \implies 0 \le \frac{n_A}{n} \le 1 \implies 0 \le f_n(A) \le 1$$

Property 2: Certainty & Impossibility

The Property

The certain event (full sample space) has probability 1; the impossible event (empty set) has probability 0.

$$P(\mathcal{S}) = 1 \quad \text{and} \quad P(\emptyset) = 0$$

Derivation

For the sure event $A = \mathcal{S}$ and the impossible event $A = \emptyset$:

$$P(\mathcal{S}) = \frac{|\mathcal{S}|}{|\mathcal{S}|} = 1 \quad \text{and} \quad P(\emptyset) = \frac{|\emptyset|}{|\mathcal{S}|} = \frac{0}{|\mathcal{S}|} = 0$$

$$f_n(\mathcal{S}) = \frac{n}{n} = 1 \quad \text{and} \quad f_n(\emptyset) = \frac{0}{n} = 0$$

Property 3: Additivity of Disjoint Events

The Property

If events A and B are **mutually exclusive** ($A \cap B = \emptyset$), then the probability of their union is the sum of their individual probabilities.

$$P(A \cup B) = P(A) + P(B)$$

Derivation

If A and B share no common elements, cardinality is additive: $|A \cup B| = |A| + |B|$. Thus,

$$\frac{|A \cup B|}{|\mathcal{S}|} = \frac{|A| + |B|}{|\mathcal{S}|} = \frac{|A|}{|\mathcal{S}|} + \frac{|B|}{|\mathcal{S}|} \implies P(A \cup B) = P(A) + P(B)$$

$$f_n(A \cup B) = \frac{n_A + n_B}{n} = \frac{n_A}{n} + \frac{n_B}{n} \implies f_n(A \cup B) = f_n(A) + f_n(B)$$

Exercises

Complement Event

Prove that the probability of the complement of any event A is given by:

$$P(A^c) = 1 - P(A)$$

Hint

Recall that A and A^c completely partition the sample space $\mathcal{S}$ without overlap.

Monotonicity

Prove that if event A is a subset of event B ($A \subseteq B$), then its probability cannot exceed the probability of B:

$$P(A) \leq P(B)$$

Hint

If $A \subseteq B$, every outcome contained in A is naturally also contained in B.

Exercises

General Addition rule

Prove that for any two events A and B (which may overlap):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Hint

First, prove principle of inclusion-exclusion for counting finite sets:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

- Explain why we must subtract the size of the intersection $|A \cap B|$.

Bridging to Formal Theory: Why Axioms?

These intuitive properties are consistent across both viewpoints:

Concept	Classical Ratio	Empirical Count
Boundedness	$0 \leq \lvert A\rvert/\lvert\mathcal{S}\rvert \leq 1$	$0 \leq n_A/n \leq 1$
Certainty	$\lvert\mathcal{S}\rvert/\lvert\mathcal{S}\rvert = 1$	$n/n = 1$
Additivity	$(\lvert A\rvert + \lvert B\rvert)/\lvert\mathcal{S}\rvert$	$(n_A + n_B)/n$

The Limitations

- **Classical definition** falls short when outcomes are not equally likely.
- **Empirical definition** relies on an impossible, unverifiable limit $(n \to \infty)$.

The Experiment

The Setup

Suppose an experimenter rolls two fair, six-sided dice.

- The fundamental sample space $\mathcal{S}$ has 36 equally likely outcomes.

The Hidden Transformation

Before you can see the dice, they are passed through a digital "black box." The box applies an **unknown function** $Z = f(x, y)$ to the outcomes of the two dice (x and y).

- You, the observer, are only shown the final output $Z \in \mathcal{T} = \{0, 1, 2, 3, 4, 5\}$.

The Core Questions

1. Can we assume that the observable outcomes in $\mathcal{T}$ are equally likely?
2. If you only have access to $\mathcal{T}$, can you mathematically calculate their probabilities?

Revealing the Hidden Function

The Truth Behind the Curtain

Let us reveal the unknown function programmed inside the black box:

$$Z = f(x, y) = |x - y| \quad \text{(The absolute difference between the dice)}$$

Calculating the Real Probabilities

We must map the outputs back to the original, equally likely 36 outcomes in $\mathcal{S}$:

- $Z = 0 \implies \{(1,1), (2,2), \ldots, (6,6)\} \implies$ 6 outcomes $\implies P(Z = 0) = \frac{6}{36}$
- $Z = 1 \implies \{(1,2), (2,1), (2,3), \ldots\} \implies$ 10 outcomes $\implies P(Z = 1) = \frac{10}{36}$
- $Z = 5 \implies \{(1,6), (6,1)\} \implies$ 2 outcomes $\implies P(Z = 5) = \frac{2}{36}$

Conclusion: They are Not Equally Likely!

The Problem

If the mapping function remains unknown, classical Calculation is Impossible

Without knowing the mapping function, it is **mathematically impossible** to calculate the true probabilities of $\mathcal{T}$ purely from its set definition.

The Alternative: Empirical Estimation

The only way to estimate the probabilities is to rely on the **empirical (relative frequency) approach**:

- Collect and record an observational stream of outcomes over n runs.
- Estimate the probability of each outcome as $f_n(Z) = \frac{n_Z}{n}$.

The Catch: An Unknown Limit

Since we can never perform infinite trials, we cannot verify if this limit truly stabilizes, leaving the mathematical reasoning incomplete!

The Problem

The Reality

Almost all real-world phenomena we try to model are exactly like this experiment. The complex systems we observe are rarely isolated, simple events.

Hidden Factors and Confounding Variables

What we measure is the cumulative output of countless underlying physical forces, behaviors, and environmental factors:

- Many of these variables are completely hidden or unknown to us.
- They interact in non-linear ways that we may not be able to model.

The Motivation for a New Approach

This complexity is precisely why we need a robust, unified definition of probability. To work around the physical limitations of classical and empirical approaches, we need a formal mathematical system that operates independently of our ability to observe every underlying factor.

Kolmogorov's Three Axioms

Let $\mathcal{S}$ be the sample space of a random experiment. To every event $A \subseteq \mathcal{S}$, we assign a real number $P(A)$ satisfying the following:

- Axiom 1: Non-Negativity For any event A, the assigned probability cannot be negative:
$$P(A) \geq 0$$
- Axiom 2: Certainty The probability of the entire sample space (the sure event) is exactly 1:
$$P(\mathcal{S}) = 1$$
- Axiom 3: Countable Additivity For any infinite sequence of mutually exclusive (disjoint) events $A_1, A_2, A_3, \ldots$ (where $A_i \cap A_j = \emptyset$ for all $i \neq j$):
$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Consequence of Axioms

Theorem: The Complement Rule

For any event A, the probability of its complement is:

$$P(A^c) = 1 - P(A)$$

Theorem: Probability of the Null Set

The probability of the empty set (impossible event) is zero:

$$P(\emptyset) = 0$$

Theorem: Monotonicity

If event A is a subset of event B ($A \subseteq B$), then:

$$P(A) \leq P(B)$$

Proof: The Complement Rule

By set definition, an event A and its complement A^c are mutually exclusive ($A \cap A^c = \emptyset$) and their union partitions the entire sample space ($A \cup A^c = \mathcal{S}$).

1. Apply **Axiom 3** (for finite disjoint events, where $A_3, A_4, \dots = \emptyset$):

$$P(A \cup A^c) = P(A) + P(A^c)$$

2. Since $A \cup A^c = \mathcal{S}$, substitute on the left side:

$$P(\mathcal{S}) = P(A) + P(A^c)$$

3. Apply **Axiom 2** ($P(\mathcal{S}) = 1$):

$$1 = P(A) + P(A^c) \implies P(A^c) = 1 - P(A) \quad \blacksquare$$

Proof: Probability of the Null Set

Let the event $A = \mathcal{S}$. Its complement is the empty set:

$$A^c = \mathcal{S}^c = \emptyset$$

1. Apply the **Complement Rule** ($P(A^c) = 1 - P(A)$) to this setup:

$$P(\emptyset) = P(\mathcal{S}^c) = 1 - P(\mathcal{S})$$

2. Now, substitute **Axiom 2** ($P(\mathcal{S}) = 1$):

$$P(\emptyset) = 1 - 1 = 0 \quad \blacksquare$$

Proof: Monotonicity

If $A \subseteq B$, we can decompose event B into two disjoint parts: the part inside A, and the part that is in B but outside A (denoted $B \cap A^c$):

$$B = A \cup (B \cap A^c) \quad \text{where} \quad A \cap (B \cap A^c) = \emptyset$$

1. Apply **Axiom 3** (Additivity):

$$P(B) = P(A) + P(B \cap A^c)$$

2. Apply **Axiom 1** (Non-negativity), we know $P(B \cap A^c) \geq 0$.
3. Therefore, adding a non-negative term to $P(A)$ yields:

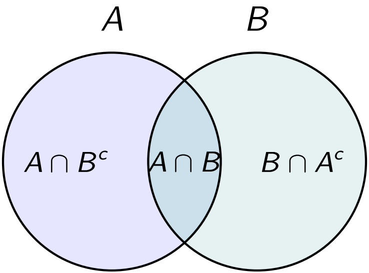
$$P(B) \geq P(A) \quad \blacksquare$$

The General Addition Rule (Overlapping Events)

Theorem: General Addition Rule

For any two events A and B (not necessarily disjoint):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Proof: General Addition Rule

We can partition the union $A \cup B$ and the event B itself into mutually exclusive parts:

1. Partition $A \cup B$: $A \cup B = A \cup (B \cap A^c)$. By Axiom 3:

$$P(A \cup B) = P(A) + P(B \cap A^c) \quad \text{— (Equation 1)}$$

2. Partition B: $B = (A \cap B) \cup (B \cap A^c)$. By Axiom 3:

$$P(B) = P(A \cap B) + P(B \cap A^c)$$

3. Solve the second equation for $P(B \cap A^c)$:

$$P(B \cap A^c) = P(B) - P(A \cap B) \quad \text{— (Equation 2)}$$

4. Substitute Equation 2 into Equation 1:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad \blacksquare$$

Exercise 1

Problem Statement

Let A and B be two events in a random experiment. You are given that:

$$P(A) = 0.7 \quad \text{and} \quad P(B) = 0.8$$

Use the axioms of probability to show that the probability of their intersection must satisfy:

$$P(A \cap B) \geq 0.5$$

Exercise 2

Problem Statement

Let A and B be any two arbitrary events (they do not have to be subsets of each other).

Prove directly from the axioms of probability that:

$$P(B \cap A^c) = P(B) - P(A \cap B)$$

Exercise 3

Problem Statement

Let E_1, E_2, and E_3 be three pairwise mutually exclusive events such that:

$$P(E_1) = 0.15, \quad P(E_2) = 0.35, \quad \text{and} \quad P(E_3) = 0.40$$

Calculate the following probabilities:
(a) $P(E_1 \cup E_2 \cup E_3)$
(b) $P(E_1^c \cap E_2^c \cap E_3^c)$

The Intuition of Conditioning

In many scenarios, we acquire partial information about the outcome of an experiment before its full result is revealed.

Restricting the Sample Space

When we learn that an event B has occurred, any outcome outside of B becomes impossible.

- The original sample space $\mathcal{S}$ collapses.
- The event B becomes our **new, effective sample space**.

We must now recalculate the probability of an event A given this restricted domain. This is denoted as $P(A \mid B)$.

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Formal Definition of Conditional Probability

Definition

Conditional Probability Let A and B be two events in a sample space $\mathcal{S}$ where $P(B) > 0$. The conditional probability of A given that B has occurred is defined as:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Immediate Operational Takeaway:

- The formula is mathematically undefined if $P(B) = 0$. We can only condition on events that have a non-zero probability of occurring.

Why is it Defined This Way? (Normalization)

We define $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$ because we want the new conditional evaluation to act as a **fully valid probability measure** over the restricted space B.

1. **Restricting to the active region:**
   - Since B is guaranteed to have occurred, any outcome in A can only occur if it lies in the intersection $A \cap B$.
2. **Establishing a new "Sure Event":**
   - If B is now our effective sample space, we must mathematically force $P(B \mid B) = 1$.
   - Division by $P(B)$ acts as a **normalizing scaling factor**:

$$P(B \mid B) = \frac{P(B \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

Properties: Satisfying Kolmogorov's Axioms

For a fixed event B where $P(B) > 0$, the conditional mapping $P(\cdot \mid B)$ satisfies all three of Kolmogorov's Axioms.

Axiom 1: Non-Negativity

For any event $A \subseteq \mathcal{S}$:

$$P(A \mid B) \geq 0$$

Proof: Since $P(A \cap B) \geq 0$ (Axiom 1) and $P(B) > 0$, their quotient must be non-negative.

Axiom 2: Certainty (Normalization)

For the entire sample space $\mathcal{S}$:

$$P(\mathcal{S} \mid B) = 1$$

Proof: Since $\mathcal{S} \cap B = B$, we have:

$$P(\mathcal{S} \mid B) = \frac{P(\mathcal{S} \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1 \quad \blacksquare$$

Properties: Countable Additivity

Axiom 3: Countable Additivity

For any infinite sequence of mutually exclusive events $A_1, A_2, \ldots$:

$$P\left(\bigcup_{i=1}^{\infty} A_i \,\middle|\, B\right) = \sum_{i=1}^{\infty} P(A_i \mid B)$$

Proof: Using distributive laws, the events $(A_i \cap B)$ are also mutually exclusive.

$$P\left(\bigcup_{i=1}^{\infty} A_i \,\middle|\, B\right) = \frac{P\left(\left(\bigcup_{i=1}^{\infty} A_i\right) \cap B\right)}{P(B)} = \frac{P\left(\bigcup_{i=1}^{\infty} (A_i \cap B)\right)}{P(B)}$$

$$= \frac{\sum_{i=1}^{\infty} P(A_i \cap B)}{P(B)} \quad \text{(by Axiom 3 on disjoint elements)}$$

$$= \sum_{i=1}^{\infty} \frac{P(A_i \cap B)}{P(B)} = \sum_{i=1}^{\infty} P(A_i \mid B) \quad \blacksquare$$

The Multiplication Rule

By rearranging the conditional probability definition, we obtain a powerful tool to find the joint probability of two overlapping events.

Theorem: The Multiplication Rule

For any two events A and B:

$$P(A \cap B) = P(B) \cdot P(A \mid B)$$

Equivalently, if we condition on A instead:

$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

This rule is highly practical for multi-stage random processes where the second stage directly depends on the outcome of the first.

Exercise: Calculating Conditional Probabilities

Problem Statement

A pair of fair, six-sided dice are rolled, yielding a sample space $\mathcal{S}$ of 36 equally likely outcomes. Let:

- Event A: The sum of the two dice is 8.
- Event B: Both dice show even numbers.

Calculate:
(a) $P(A)$ and $P(B)$
(b) The conditional probabilities $P(A \mid B)$ and $P(B \mid A)$

Extending the Multiplication Rule (The Chain Rule)

The multiplication rule generalizes naturally to three or more dependent events.

Theorem: The General Chain Rule

For any sequence of events $A_1, A_2, \ldots, A_n$:

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1) \cdot P(A_2 \mid A_1) \cdot P(A_3 \mid A_1 \cap A_2) \cdots P(A_n \mid \bigcap_{i=1}^{n-1} A_i)$$

Example: Sequential Drawing Without Replacement

- Suppose we draw 3 cards sequentially from a 52-card deck without replacement.
- Let A_i be the event of drawing an Ace on the i-th draw:

$$P(A_1 \cap A_2 \cap A_3) = \frac{4}{52} \cdot \frac{3}{51} \cdot \frac{2}{50}$$

Exercise: Multi-stage Dependent Events

Problem Statement

An urn contains 5 red balls and 4 blue balls. Three balls are drawn sequentially, one by one, **without replacement**.

Find the probability that the balls are drawn in the exact sequence:

$$\text{First: Red} \quad \rightarrow \quad \text{Second: Blue} \quad \rightarrow \quad \text{Third: Red}$$

Hint: Let R_i and B_i denote drawing a Red or Blue ball on draw i. Apply the Chain Rule.

Independence of Events

Two events are independent if the occurrence of one provides absolutely no informational update regarding the likelihood of the other.

Definition

Statistical Independence Two events A and B are said to be independent if and only if:

$$P(A \cap B) = P(A) \cdot P(B)$$

Informational Intuition:

- Knowing that B happened does not change the probability of A:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B)}{P(B)} = P(A)$$

- Similarly, knowing A happened does not change the likelihood of B:

$$P(B \mid A) = P(B)$$

Exercise:

Problem Statement

Let A and B be two events such that:

$$P(A) = 0.4, \quad P(B) = 0.5, \quad P(A \cup B) = 0.7$$

Are the events A and B independent? Justify your answer using the definition of independence.

Problem Statement

Suppose A and B are two mutually exclusive (disjoint) events, and $P(A) > 0$ and $P(B) > 0$. Are A and B independent? Explain why or why not.

Exercise :

Problem Statement

Consider rolling two fair, six-sided dice. Let:

- Event A: The first die shows an odd number $\{1, 3, 5\}$.
- Event B: The sum of the two dice is odd.

Intuitively, one might suspect these events are dependent because the sum depends on both dice. Show that A and B are actually **independent**.

Can you explain this logically without any calculations?

Reversing Conditional Probabilities

In many applications, we know the conditional probability of an effect given a cause, but we actually observe the effect and want to deduce the probability of the cause.

The Problem of Inversion

- We easily know $P(\text{Symptom} \mid \text{Disease})$.
- But a patient walks in with the symptom; we must find $P(\text{Disease} \mid \text{Symptom})$.
- How do we mathematically reverse conditioning to find $P(B \mid A)$ if we only have $P(A \mid B)$?

The Base Case: Two Exhaustive Scenarios

Let us start with the simplest possible scenario: an event B either happens, or it does not (B^c). These two events are **mutually exclusive** ($B \cap B^c = \emptyset$) and **exhaustive** ($B \cup B^c = \mathcal{S}$).

For any target event A, we can decompose it into these two scenarios:

$$A = (A \cap B) \cup (A \cap B^c)$$

Since these two components are disjoint, their probabilities add:

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

Applying the multiplication rule gives the **Simple Law of Total Probability**:

$$P(A) = P(A \mid B)P(B) + P(A \mid B^c)P(B^c)$$

Bayes' Theorem (Simple Form)

Using our definition of conditional probability, we can now write the probability of B given that we observe event A:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

By replacing the numerator with the multiplication rule, and the denominator with our simple Law of Total Probability, we get:

Theorem

Bayes' Theorem (Two-Event Form) For any two events A and B where $P(A) > 0$ and $0 < P(B) < 1$:

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A \mid B)P(B) + P(A \mid B^c)P(B^c)}$$

Scaling Up: General Partitions

What if there are more than just two underlying scenarios? We can scale this logic up to any arbitrary number of mutually exclusive possibilities.

Definition : A collection of events $B_1, B_2, \ldots, B_n$ is a **partition** of the sample space $\mathcal{S}$ if:

1. They are pairwise mutually exclusive:

$$B_i \cap B_j = \emptyset \quad \text{for all } i \neq j$$

2. Their union completely exhausts the sample space:

$$\bigcup_{i=1}^{n} B_i = \mathcal{S}$$

3. Each partition slice has a positive probability: $P(B_i) > 0$.

The General Law of Total Probability

When an event A can occur under several distinct, disjoint scenarios $B_1, \ldots, B_n$, we find its total probability by summing across all partitions.

Theorem

The Law of Total Probability (General Form) Let $B_1, B_2, \ldots, B_n$ be a partition of $\mathcal{S}$. For any event A:

$$P(A) = \sum_{i=1}^{n} P(A \cap B_i)$$

Applying the multiplication rule, this is equivalent to:

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i) P(B_i)$$

General Form of Bayes' Theorem

Combining the conditional probability definition with the general Law of Total Probability allows us to resolve the posterior probability for any specific scenario B_k.

Theorem

Bayes' Theorem (General Form) Let $B_1, B_2, \ldots, B_n$ be a partition of the sample space $\mathcal{S}$. For any event A where $P(A) > 0$, the posterior probability for any specific slice B_k is:

$$P(B_k \mid A) = \frac{P(A \mid B_k)P(B_k)}{\sum_{i=1}^{n} P(A \mid B_i)P(B_i)}$$

The denominator is simply the total probability $P(A)$ expanded across all n possible options.

Anatomy of the Bayesian Terms

Each term in Bayes' Theorem plays a distinct role in statistical inference and updating beliefs.

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}$$

- **Prior Probability $P(B)$:** Our initial assessment of the likelihood of event B before observing the new evidence.
- **Likelihood $P(A \mid B)$:** The probability that the evidence A would occur given that scenario B is true.
- **Posterior Probability $P(B \mid A)$:** The updated probability of B *after* taking the evidence A into account.
- **Evidence $P(A)$:** The overall probability of observing the evidence A across all possible scenarios.

Exercise: Medical Diagnostics (Base Rate Fallacy)

Problem Statement

A rare medical condition affects exactly 1% of a population. A diagnostic test is developed to screen for this condition:

- **Sensitivity** (True Positive Rate): If a person has the condition, the test yields a positive result $(+)$ with probability 0.99.
- **False Positive Rate**: If a person does not have the condition, the test still mistakenly yields a positive result $(+)$ with probability 0.02.

If a randomly selected person from the population tests positive $(+)$, what is the probability that they actually have the condition?

Define your events clearly: Let D be the event of having the disease, and let T^+ be the event of testing positive.

Exercise 2: Spam Notifications

Problem Statement

A digital user receives notifications across three primary channels: Email (E), SMS (S), and Instant Messaging (M):

- 50% of all notifications arrive via Email, 30% via SMS, and 20% via Instant Message.
- The likelihood of a notification containing a harmful link (L) differs by channel:
  - For Emails, only 1% contain harmful links.
  - For SMS, 4% contain harmful links.
  - For Instant Messages, 10% contain harmful links.

If a randomly received notification is flagged as containing a harmful link (L), what is the probability that it arrived via Instant Messaging (M)?

Why Do We Need Random Variables?

Up to now, we have defined events as qualitative subsets of a sample space:

$$\mathcal{S} = \{\text{Heads}, \text{Tails}\} \quad \text{or} \quad \mathcal{S} = \{\text{Red}, \text{Blue}, \text{Green}\}$$

The Limitation of Qualitative Outcomes

- We cannot perform algebraic operations (like addition, averaging, or scaling) on raw outcomes like "Heads" or "Red".
- To build mathematical models, calculate averages, or measure variances, we must translate these outcomes into numerical values.

We need a systematic rule that assigns a real number to every single outcome in our sample space.

Formal Definition of a Random Variable

Definition

Random Variable Let $\mathcal{S}$ be the sample space of a random experiment. A **random variable** X is a real-valued function that maps every outcome $s \in \mathcal{S}$ to a unique real number $X(s) \in \mathbb{R}$.

$$X : \mathcal{S} \to \mathbb{R}$$

Crucial Conceptual Caveat

Despite its name, a random variable is **not a variable** and is **not random** in the traditional sense.

- It is a deterministic, mathematically defined **function**.
- The "randomness" comes solely from the underlying experiment which determines which outcome $s \in \mathcal{S}$ occurs.

The Range of a Random Variable

The collection of all numerical values that a random variable can actually take is called its **range** or **support**, often denoted by $\mathcal{T}$ or $X(\mathcal{S})$.

$$\mathcal{T} = \{x \in \mathbb{R} \mid X(s) = x \text{ for some } s \in \mathcal{S}\}$$

Example: Tossing Two Fair Coins

Let the sample space be:

$$\mathcal{S} = \{HH, HT, TH, TT\}$$

Let X be the random variable representing the **number of Heads**.

- $X(HH) = 2$
- $X(HT) = 1$
- $X(TH) = 1$
- $X(TT) = 0$

The support of X is the discrete set: $\mathcal{T} = \{0, 1, 2\}$.

Discrete vs. Continuous Random Variables

Depending on the nature of their support set $\mathcal{T}$, random variables are classified into two fundamental domains:

1. Discrete Random Variables

A random variable is **discrete** if its range $\mathcal{T}$ consists of a finite or countably infinite set of values (such as integers).

- Outcomes can be listed in a sequence: $\{x_1, x_2, x_3, \dots\}$.

2. Continuous Random Variables

A random variable is **continuous** if its range $\mathcal{T}$ consists of an uncountable interval or union of intervals of real numbers.

- It can assume any real-valued measurement in its range.

Discrete Example: Rolling Until Success

Consider rolling a fair, six-sided die repeatedly until you roll a "6". Let Y be the random variable representing the **total number of rolls required**.

Defining the Support Set

The physical experiment could theoretically go on forever if you keep rolling non-sixes.

- If you roll a 6 on the very first try: $Y = 1$.
- If you roll a 2, then a 6: $Y = 2$.
- If you roll a 4, 3, 5, then a 6: $Y = 4$.

Because we can count the trials, the support set $\mathcal{T}$ is a countably infinite set of positive integers:

$$\mathcal{T} = \{1, 2, 3, 4, 5, \ldots\}$$

Continuous Example: Spin the Needle

Consider a circular dial with a single spinning needle. The circumference of the dial is precisely calibrated to be 1 meter, starting at a marked line (0).

Let the needle be spun. Let Z be the random variable representing the **distance (in meters) along the circumference from 0 to the stopped needle**.

Defining the Support Set

Since physical space is continuous, the pointer can stop at any fractional decimal point along the dial (e.g., 0.4578... meters).

- Minimum possible value: 0 (stopping exactly on the starting line).
- Maximum possible value: approaching 1 (but resetting to 0 if it completes the loop).

The range of Z is represented as an uncountable interval:

$$\mathcal{T} = [0, 1)$$

Discrete Example: Rolling Two Dice

Suppose two fair, six-sided dice are rolled, yielding a sample space $\mathcal{S}$ of 36 equally likely outcomes.

Let Y be the random variable representing the **sum of the two dice**:

$$Y(i,j) = i + j \quad \text{where } (i,j) \in \mathcal{S}$$

- The minimum possible value is: $Y(1,1) = 2$.
- The maximum possible value is: $Y(6,6) = 12$.
- The range is the discrete list of values:

$$\mathcal{T} = \{2,3,4,5,6,7,8,9,10,11,12\}$$

Since the range is finite, Y is a **discrete random variable**.

Continuous Example: Device Lifetime

Consider an experiment where we observe the operational lifetime (in hours) of a continuous electronic component until it experiences a structural failure.

The physical sample space consists of all non-negative real numbers:

$$\mathcal{S} = \{t \in \mathbb{R} \mid t \geq 0\}$$

Let T be the random variable representing the **lifetime of the component**:

$$T(t) = t \quad \text{for any } t \in \mathcal{S}$$

- Since the time of failure can be any fractional value (e.g., 104.532 hours), the range is the continuous interval:

$$\mathcal{T} = [0, \infty)$$

Because this range is uncountable, T is a **continuous random variable**.

Exercise: Classifying Random Variables

For each of the following scenarios, identify:

1. The random variable X of interest.
2. Whether X is **discrete** or **continuous**.
3. The mathematically defined range/support $\mathcal{T}$ of X.

Scenarios

- **Scenario A:** A fair coin is tossed repeatedly until the first "Heads" appears. We count the total number of tosses required.
- **Scenario B:** A researcher measures the precise blood pressure level (in mmHg) of a patient.
- **Scenario C:** A commuter records the exact waiting time (in minutes) for a city bus, which is guaranteed to arrive within 30 minutes.

The Induced Probability Map

A random variable $X : \mathcal{S} \to \mathbb{R}$ is a completely deterministic, algebraic function. It does not possess any "inherent" randomness.

Where does the probability come from?

The probability of X taking on a specific value is entirely **derived** from the probability of the underlying outcomes in the sample space $\mathcal{S}$ that map to that value.

If the underlying physical experiment is performed and yields an outcome $s \in \mathcal{S}$, the random variable evaluates to the number $X(s)$.

Translating Events to Real Numbers

Let x be a real number. We want to define what $P(X = x)$ means.

We look backward at the sample space $\mathcal{S}$ and gather all outcomes s that map to x. We call this subset the **pre-image** of x:

$$E_x = \{s \in \mathcal{S} \mid X(s) = x\}$$

Definition

The Probability Law of X The probability that the random variable X equals x is defined as the probability of its underlying event E_x occurring:

$$P(X = x) \triangleq P(\{s \in \mathcal{S} \mid X(s) = x\})$$

This is often written in compact shorthand as:

$$P(X = x) = P(s \in \mathcal{S} : X(s) = x)$$

Example: Tossing Two Fair Coins

Let $\mathcal{S} = \{HH, HT, TH, TT\}$ with equally likely outcomes ($P(s) = 1/4$). Let X represent the **number of Heads**:

$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0$$

To find $P(X = 1)$, we identify which outcomes in $\mathcal{S}$ map to 1:

$$E_1 = \{s \in \mathcal{S} \mid X(s) = 1\} = \{HT, TH\}$$

Since E_1 is a standard, well-defined event in our sample space, we can calculate its probability:

$$P(X = 1) = P(E_1) = P(\{HT, TH\}) = \frac{2}{4} = 0.5$$

The number 1 is assigned the collective probability mass of the outcomes HT and TH.

Extending to Intervals and Ranges

We do not just calculate probability at single points. We often want to know the probability of a range, such as $P(a \leq X \leq b)$ or $P(X \leq x)$.

The exact same **mapping** rule applies! We collect all outcomes in the sample space $\mathcal{S}$ that map into that interval:

$$P(a \leq X \leq b) \triangleq P(\{s \in \mathcal{S} \mid a \leq X(s) \leq b\})$$

Shorthand Notation Caveat

When you write $P(X \leq x)$, you are using a convenient shorthand.
The mathematical reality behind that shorthand is:

$$P(X \leq x) = P(\{s \in \mathcal{S} \mid X(s) \leq x\})$$

This translation connects the real number world $(X \leq x)$ back to the set theory world $(\mathcal{S})$ where Kolmogorov's Axioms live!

Exercise 1: Discrete Pre-image Mapping

Consider rolling a fair die twice. The sample space $\mathcal{S}$ contains 36 equally likely outcomes.

Let the random variable X be the **maximum of the two rolls**:

$$X(i, j) = \max(i, j) \quad \text{for } (i, j) \in \mathcal{S}$$

Your Task

1. Identify the pre-image subset $E_3 = \{s \in \mathcal{S} \mid X(s) = 3\}$.
2. Use the probability law of X to calculate $P(X = 3)$.

Exercise 2: Continuous Range Translation

Let the sample space be the interval $\mathcal{S} = [0, 10]$. For any sub-interval $[a, b] \subseteq [0, 10]$, the probability is proportional to its length:

$$P([a, b]) = \frac{b - a}{10}$$

Let the random variable Y be defined by the linear mapping:

$$Y(s) = 2s + 3 \quad \text{for } s \in \mathcal{S}$$

Your Task

1. Translate the event $\{Y \leq 11\}$ back to its corresponding pre-image subset in the sample space $\mathcal{S}$.
2. Calculate the probability $P(Y \leq 11)$.

Exercise 3: Cloud Database Reliability

A cloud database replicates data across 3 independent servers. The sample space $\mathcal{S}$ represents the state of each server: (x_1, x_2, x_3) where $x_i \in \{0, 1\}$ ($1 =$ online, $0 =$ offline). The probability of any individual server failing (offline) is 0.1. The servers operate independently.
Let the random variable C represent the **available system throughput** (in Gbps), mapped as a function of the number of online servers $k = x_1 + x_2 + x_3$:

$$C(s) = \begin{cases} 100 & \text{if } k = 3 \\ 50 & \text{if } k = 2 \\ 20 & \text{if } k = 1 \\ 0 & \text{if } k = 0 \end{cases}$$

Calculate the probability $P(C \geq 50)$.

Exercise 4: Reactor Pressure Telemetry

Consider an industrial pressure sensor placed in a chemical reactor. The operational pressure s (in psi) ranges uniformly over the sample space $\mathcal{S} = [0, 200]$ psi, where $P([a, b]) = \frac{b-a}{200}$ for any sub-interval.
The safety telemetry system converts the raw pressure s to a safety index Y using a piecewise logarithmic mapping:

$$Y(s) = \begin{cases} 0 & \text{if } 0 \leq s < 1 \quad \text{(noise floor)} \\ 20\log_{10}(s) & \text{if } 1 \leq s \leq 100 \quad \text{(telemetry monitoring)} \\ 40 & \text{if } 100 < s \leq 200 \quad \text{(saturation limit)} \end{cases}$$

Your Task

1. Translate the telemetry reading event $\{20 \leq Y \leq 40\}$ back to its corresponding pre-image subset $E \subseteq \mathcal{S}$.
2. Calculate the probability $P(20 \leq Y \leq 40)$ using the probability law.

What is a "Distribution"?

A random variable X maps outcomes from a sample space $\mathcal{S}$ to the real numbers $\mathbb{R}$.
The way the probability mass is scattered, allocated, or "distributed" over these real numbers is called the **probability distribution** of X.

To study distributions systematically without invoking measure theory, we use specific functions:

- **The Cumulative Distribution Function (CDF)** — Applies universally to all random variables.
- **The Probability Mass Function (PMF)** — Specifically for discrete random variables.
- **The Probability Density Function (PDF)** — Specifically for continuous random variables.

Why Do We Need a Distribution Function?

Previously, calculating probabilities required looking backward to the sample space:

$$P(X \in B) = P(\{s \in \mathcal{S} \mid X(s) \in B\})$$

While mathematically sound, this approach has severe practical limitations:

- It is highly inefficient to map back to the sample space $\mathcal{S}$ for every single probability calculation.
- In many real-world applications, the original sample space $\mathcal{S}$ is complex, high-dimensional, or completely hidden from us.

The Solution

A **distribution function** acts as a mathematical proxy. Once we establish this function on the real line $\mathbb{R}$:

- **All** probabilities associated with X can be calculated using this function **alone**.

The Cumulative Distribution Function (CDF)

The most fundamental and universal way to describe the distribution of any random variable is through cumulative probabilities.

Definition

Cumulative Distribution Function Let X be a random variable. The **Cumulative Distribution Function (CDF)** of X, denoted by $F(x)$ (or $F_X(x)$), is defined for any real number x as:

$$F(x) \triangleq P(X \leq x) \quad \text{for } x \in \mathbb{R}$$

Key Concept

The value $F(x)$ represents the accumulated probability of all outcomes in the sample space that map to a value less than or equal to x.

Why $(-\infty, x]$ is Enough: Reconstructing Events

The CDF is defined strictly using semi-infinite intervals:

$$F(a) \triangleq P(X \leq x) = P(X \in (-\infty, x])$$

A Fundamental Result

Every "nice" or mathematically well-behaved subset of the real numbers can be constructed from intervals of the form $(-\infty, x]$ using basic set operations.

Consequently, the probability of any reasonable event can be written as an addition or subtraction combination of the distribution function values:

Properties of the CDF

The CDF $F(x)$ is a mathematically well-behaved function. For any random variable X, its CDF satisfies the following structural properties:

1. **Monotonicity:** $F(x)$ is a non-decreasing function:

$$\text{If } a \leq b, \text{ then } F(a) \leq F(b)$$

2. **Asymptotic Limits:**

$$\lim_{x \to \infty} F(x) = 1 \quad \text{and} \quad \lim_{x \to -\infty} F(x) = 0$$

3. **Right-Continuity:** $F(x)$ is continuous from the right at every point:

$$\lim_{t \to x^+} F(t) = F(x)$$

Discrete Case: Probability Mass Function (PMF)

For discrete random variables (which take on countable values), we define probability at exact, individual points.

Definition

Probability Mass Function Let X be a discrete random variable with range $\mathcal{T}$. The **Probability Mass Function (PMF)** of X, denoted by $p(x)$ (or $p_X(x)$), is:

$$p(x) \triangleq P(X = x) \quad \text{for } x \in \mathbb{R}$$

Fundamental Axiomatic Properties

Every valid PMF must satisfy:

- $p(x) \geq 0$ for all x.
- $\sum_{x \in \mathcal{T}} p(x) = 1$.

Continuous Case: Probability Density Function (PDF)

For continuous random variables, the probability of taking on any single exact value is zero ($P(X = x) = 0$). Instead, we describe the probability over an interval using density.

Definition

Probability Density Function Let X be a continuous random variable. A non-negative, integrable function $f(x)$ is the **Probability Density Function (PDF)** of X if for any interval $[a, b]$:

$$P(a \leq X \leq b) = \int_a^b f(x)\, dx$$

Axiomatic Constraints

- $f(x) \geq 0$ for all $x \in \mathbb{R}$.
- $\int_{-\infty}^{\infty} f(x)\, dx = F(\infty) = 1$.

Exercise: Verifying a Valid PMF

Suppose a system database query returns a variable number of matching records, X. The probability mass function of X is modeled by:

$$p(x) = \begin{cases} k \cdot x^2 & \text{if } x \in \{1, 2, 3, 4\} \\ 0 & \text{otherwise} \end{cases}$$

Your Task

1. Find the value of the normalization constant k such that $p(x)$ represents a mathematically valid PMF.
2. Determine and write down the Cumulative Distribution Function (CDF) $F(x)$ for this discrete random variable.

Exercise: Verifying a Valid PDF

Consider a continuous random variable X representing the time (in hours) a user spends actively on a platform in a session. The density is modeled by:

$$f(x) = \begin{cases} c \cdot (2x - x^2) & \text{if } x \in [0, 2] \\ 0 & \text{otherwise} \end{cases}$$

Your Task

Find the value of the normalization constant c such that $f(x)$ represents a mathematically valid PDF, and determine the CDF $F(x)$.

Expectation: The Center of Mass

The **mathematical expectation** (or mean) of a random variable represents its long-run average value over infinitely repeated trials.

Physically, it acts as the **center of gravity** (centroid) of the probability distribution.

Definition

Expected Value (Discrete) Let X be a discrete random variable with support set $\mathcal{T}$ and probability mass function $p(x) = P(X = x)$. The expected value of X, denoted $E[X]$ or μ, is:

$$E[X] \triangleq \sum_{x \in \mathcal{T}} x \cdot p(x)$$

This sum is defined provided that it is absolutely convergent, i.e., $\sum |x| p(x) < \infty$.

Expectation: Continuous Analogue

For a continuous random variable, we cannot sum over individual points because any single point has zero probability.
Instead, we integrate over the continuous support using the probability density function (PDF).

Definition

Expected Value (Continuous) Let X be a continuous random variable with probability density function $f(x)$. The expected value of X is defined as:

$$E[X] \triangleq \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

This integral is defined provided that it is absolutely convergent, i.e., $\int |x| f(x)\, dx < \infty$.

Linearity of Expectation

Expectation is a **linear operator**. This means it preserves scaling and addition, regardless of whether the random variables are dependent or independent.

Theorem

Linearity Properties Let X be a random variable, and let a and b be constants. Then:

$$E[aX + b] = aE[X] + b$$

Proof: Linearity of Expectation

We prove this for the discrete case. The continuous proof follows identically by replacing summations with integrals.
Let X have support $\mathcal{T}$ and PMF $p(x)$. Apply the definition of expectation to $g(X) = aX + b$:

$$\begin{aligned} E[aX+b] &= \sum_{x\in\mathcal{T}} (ax+b)\cdot p(x) \\ &= \sum_{x\in\mathcal{T}} (ax\cdot p(x) + b\cdot p(x)) \\ &= \sum_{x\in\mathcal{T}} ax\cdot p(x) + \sum_{x\in\mathcal{T}} b\cdot p(x) \end{aligned}$$

Pulling the constants out of the summations yields:

$$E[aX+b] = a\sum_{x\in\mathcal{T}} x\cdot p(x) + b\sum_{x\in\mathcal{T}} p(x)$$

Applying $E[X] = \sum xp(x)$ and Axiom 2 ($\sum p(x) = 1$):

$$E[aX+b] = aE[X] + b\cdot(1) = aE[X] + b \quad \blacksquare$$

Variance: Quantifying Spread

While the mean tells us where the distribution is centered, it tells us nothing about how spread out the values are.

Variance measures the average squared deviation of the random variable from its mean (μ).

Definition

Variance Let X be a random variable with mean $\mu = E[X]$. The variance of X, denoted $Var(X)$ or σ^2, is:

$$Var(X) \triangleq E[(X-\mu)^2]$$

Standard Deviation

Because variance is measured in squared units, we often use the **Standard Deviation** (σ) to keep the units consistent with the original data:

Computational Formula for Variance

Evaluating the definition $E[(X-\mu)^2]$ directly can be algebraically tedious. There exists a much simpler, standard computational formula.

Theorem

Computational Identity For any random variable X:

$$Var(X) = E[X^2] - (E[X])^2$$

Derivation: Computational Formula

Let $\mu = E[X]$. Expanding the squared term inside the expectation definition:

$$Var(X) = E[(X-\mu)^2] = E[X^2 - 2\mu X + \mu^2]$$

Now, apply the linearity of expectation to distribute the operator across each term:

$$Var(X) = E[X^2] - E[2\mu X] + E[\mu^2]$$

Since μ is a constant, we can pull it out of the expectation. Also, the expectation of a constant is the constant itself ($E[\mu^2] = \mu^2$):

$$Var(X) = E[X^2] - 2\mu E[X] + \mu^2$$

Substitute $E[X] = \mu$ back into the expression:

$$Var(X) = E[X^2] - 2\mu(\mu) + \mu^2 = E[X^2] - 2\mu^2 + \mu^2$$

$$Var(X) = E[X^2] - \mu^2 = E[X^2] - (E[X])^2 \quad \blacksquare$$

Properties of Variance: Scaling and Shifts

Unlike expectation, variance is **not** a linear operator. Adding a constant shift does not change the spread of the data, while scaling a variable scales its variance quadratically.

Theorem

Variance Transformations Let X be a random variable, and let a and b be constants. Then:

$$Var(aX + b) = a^2 Var(X)$$

Proof: Variance Transformations

Let $Y = aX + b$. First, find the mean of the transformed variable Y:

$$E[Y] = E[aX + b] = aE[X] + b$$

Now, substitute Y and $E[Y]$ into the definition of variance:

$$\begin{aligned} Var(aX + b) &= E\left[((aX + b) - E[aX + b])^2\right] \\ &= E\left[(aX + b - (aE[X] + b))^2\right] \end{aligned}$$

Notice that the constant shift b cancels out completely:

$$\begin{aligned} Var(aX + b) &= E\left[(aX - aE[X])^2\right] \\ &= E\left[(a(X - E[X]))^2\right] \\ &= E\left[a^2(X - E[X])^2\right] \end{aligned}$$

Pulling the constant factor a^2 out of the expectation:

$$Var(aX + b) = a^2E\left[(X - E[X])^2\right] = a^2Var(X) \quad \blacksquare$$

Exercise 1: Server Latency Cost Mapping

An API server experiences variable request latency. Let X represent the processing latency (in milliseconds) with the following PMF:

x	10 ms	50 ms	100 ms
$p(x)$	0.6	0.3	0.1

The operating cost (in micro-dollars) of processing a request of latency X is determined by the quadratic relationship:

$$C(X) = 2X^2 + 10$$

Your Task

Calculate the expected processing cost $E[C(X)]$ per request.

Exercise 2: Precision Sensor Noise

An industrial analog temperature sensor outputs a voltage reading V. Due to thermal interference, the sensor error $X = V - V_{\text{true}}$ (in millivolts) is a continuous random variable distributed uniformly on the range $[-5, 5]$ mV, with a constant probability density:

$$f(x) = \frac{1}{10} \quad \text{for } x \in [-5, 5]$$

Your Task

1. Compute the expected calibration error $E[X]$.
2. Compute the variance $Var(X)$ to quantify the sensor's precision limit.

Central Tendency: The Mathematical Mean

The expected value $E[X]$ we defined is physically known as the **mean** (μ). It is the most common measure of **central tendency**.

Key Characteristics of the Mean

- **Physical Centroid:** It acts as the exact balancing point of the probability mass curve or histogram.
- **Sensitivity:** Because its calculation factors in every value weighted by its probability, the mean is **highly sensitive to outliers** (skewed tails).
- **Linear Balance:** The sum (or integral) of deviations from the mean is always exactly zero:

$$E[X - \mu] = 0$$

Alternative Measures: The Median

When a distribution has extreme values or heavy asymmetry, the mean may not reflect the "typical" value. In such cases, we use the **median**.

Definition

The Median (m) The median m of a random variable X is a real number that divides the probability distribution into two equal halves.

- **Continuous Case:** m is the value satisfying:

$$F(m) = \int_{-\infty}^{m} f(x)\,dx = 0.5$$

- **Discrete Case:** m is any value satisfying:

$$P(X \leq m) \geq 0.5 \quad \text{and} \quad P(X \geq m) \geq 0.5$$

Core Advantage: The median is **robust**; changing extreme values has absolutely zero effect on its position.

Alternative Measures: The Mode

Another fundamental measure of central tendency is the **mode**, which indicates the most likely or popular outcome.

Definition

The Mode (x^*) The mode x^* is the value of x where the probability distribution reaches its maximum concentration or peak.

- **Discrete Case:** The point x^* that maximizes the PMF:

$$p(x^*) \geq p(x) \quad \text{for all } x$$

- **Continuous Case:** The point x^* that maximizes the density function $f(x)$ (typically found where $f'(x^*) = 0$ and $f''(x^*) < 0$).

Distribution Shapes

Unlike mean and median, a distribution can be **unimodal** (one peak), **bimodal** (two peaks), or **multimodal** (several peaks).

Comparing Central Tendency: Symmetry vs. Skewness

The relative positions of the Mean, Median, and Mode reveal the geometric **skewness** (asymmetry) of a distribution.

1. Perfectly Symmetric Distributions

The peaks, physical balance points, and 50% splitting lines align perfectly:

$$\text{Mean} = \text{Median} = \text{Mode}$$

2. Skewed Distributions

- **Right-Skewed (Positive):** A long right tail pulls the mean upward. The mode remains at the main peak, with the median sitting safely in between:

$$\text{Mode} < \text{Median} < \text{Mean}$$

- **Left-Skewed (Negative):** A long left tail pulls the mean downward:

$$\text{Mean} < \text{Median} < \text{Mode}$$

Robust Variability: Median Absolute Deviation

Just as the mean has a robust counterpart (the median), variance has a robust alternative known as the **Median Absolute Deviation (MAD)**. Variance and standard deviation are highly sensitive to extreme outliers because they square deviations. MAD resolves this.

Definition

Median Absolute Deviation (MAD) Let X be a random variable with median m. The Median Absolute Deviation of X is defined as:

$$MAD \triangleq \text{median}(|X - m|)$$

Why Use MAD?

- **Extreme Outlier Resistance:** A single outlier can inflate variance infinitely, whereas it barely changes the MAD.
- **Preserves Scale:** It measures variability in the same units as the original data without squaring the errors.

Exercise 3: Comparing Central Tendency Measures

Consider a continuous random variable X with the following bounded probability density function:

$$f(x) = \begin{cases} 3x^2 & \text{if } x \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

Your Task

1. Find the expected value/mean (μ).
2. Calculate the median (m).
3. Determine the mode (x^*).
4. Compare these three values to identify the geometric skewness of this distribution.

Why Do We Need Joint Distributions?

In most real-world scenarios, an experiment produces multiple numerical outcomes simultaneously.
To model relationships, associations, and dependencies, we must look at variables collectively rather than in isolation.

Key Objectives

- Learn to describe the joint behavior of multiple random variables.
- Understand how to extract individual probability laws from joint laws.
- Quantify the strength and direction of linear relationships between variables.

Discrete Case: Joint Probability Mass Function

When X and Y are both discrete random variables, we describe their joint probabilities using a single function of two variables.

Definition

Joint Probability Mass Function The joint probability mass function (joint PMF) of discrete random variables X and Y is defined for all real numbers x and y as:

$$p(x, y) \triangleq P(X = x, Y = y)$$

Axiomatic Properties

Every valid joint PMF must satisfy:

- $p(x, y) \geq 0$ for all (x, y).
- $\sum_x \sum_y p(x, y) = 1$.

Continuous Case: Joint Probability Density Function

When variables are continuous, probability at a single point is zero. Instead, we define probability over regions using a joint density surface.

Definition

Joint Probability Density Function The joint probability density function (joint PDF) of continuous random variables X and Y is a function $f(x, y)$ such that for any region $A \subseteq \mathbb{R}^2$:

$$P((X, Y) \in A) = \iint_A f(x, y)\, dx\, dy$$

Required Constraints

- $f(x, y) \geq 0$ for all $(x, y) \in \mathbb{R}^2$.
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$.

Extracting Individual Laws: Marginal Distributions

If we are given a joint distribution, we can recover the individual probability law of X or Y by summing or integrating out the other variable.

Marginal PMF (Discrete)

We sum across all possible values of the unwanted variable:

$$p_X(x) = \sum_y p(x, y) \quad \text{and} \quad p_Y(y) = \sum_x p(x, y)$$

Marginal PDF (Continuous)

We integrate across the entire domain of the unwanted variable:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy \quad \text{and} \quad f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$$

Independence of Random Variables

We previously defined independence for events. Similarly, two random variables are independent if knowing the value of one provides zero information about the other.

Definition

Statistical Independence Random variables X and Y are independent if their joint distribution factors cleanly into the product of their individual marginal distributions.

- **Discrete Case:**

$$p(x, y) = p_X(x) \cdot p_Y(y) \quad \text{for all } x, y$$

- **Continuous Case:**

$$f(x, y) = f_X(x) \cdot f_Y(y) \quad \text{for all } x, y$$

Covariance: Quantifying Relationship Direction

While expectation tells us about the center of individual variables, we need a metric to evaluate how they vary together.

Definition

Covariance Let X and Y be random variables with means $\mu_X = E[X]$ and $\mu_Y = E[Y]$. The covariance of X and Y, denoted $Cov(X, Y)$, is:

$$Cov(X, Y) \triangleq E[(X - \mu_X)(Y - \mu_Y)]$$

Intuitive Interpretation

- **Positive Covariance:** When X is above its mean, Y tends to be above its mean (positive linear trend).
- **Negative Covariance:** When X is above its mean, Y tends to be below its mean (inverse linear trend).
- **Zero Covariance:** No linear association.

Computational Identity for Covariance

Evaluating the definition $E[(X - \mu_X)(Y - \mu_Y)]$ directly requires calculating deviations for every point, which is algebraically tedious. Instead, we use a much simpler standard identity.

Theorem

Computational Formula For any two random variables X and Y:

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

Let us derive this relationship on the next slide using the properties of expectation.

Derivation: Computational Formula

Let $\mu_X = E[X]$ and $\mu_Y = E[Y]$. Expanding the terms inside the expectation definition:

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

$$Cov(X, Y) = E[XY - \mu_Y X - \mu_X Y + \mu_X \mu_Y]$$

Apply the linearity of expectation to distribute the operator across each term:

$$Cov(X, Y) = E[XY] - E[\mu_Y X] - E[\mu_X Y] + E[\mu_X \mu_Y]$$

Since the means μ_X and μ_Y are constants, we pull them out of the expectation:

$$Cov(X, Y) = E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y$$

Substitute $E[X] = \mu_X$ and $E[Y] = \mu_Y$ back into the expression:

$$Cov(X, Y) = E[XY] - \mu_Y \mu_X - \mu_X \mu_Y + \mu_X \mu_Y$$

$$Cov(X, Y) = E[XY] - \mu_X \mu_Y = E[XY] - E[X]E[Y] \quad \blacksquare$$

Properties of Covariance

Covariance behaves according to the following mathematical rules:

Core Covariance Properties

Let X, Y, and Z be random variables, and let a and b be constants.

1. **Symmetry:**
$$Cov(X, Y) = Cov(Y, X)$$

2. **Covariance with Self is Variance:**
$$Cov(X, X) = Var(X)$$

3. **Scaling and Bilinearity:**
$$Cov(aX, bY) = ab\, Cov(X, Y)$$

4. **Additivity:**
$$Cov(X + Y, Z) = Cov(X, Z) + Cov(Y, Z)$$

Independence and Covariance

What happens to covariance if the random variables are independent?

Theorem

Covariance of Independent Variables If X and Y are independent random variables, then:

$$Cov(X, Y) = 0$$

Proof Sketch

If X and Y are independent, then their joint expectation factors: $E[XY] = E[X]E[Y]$. Substituting this into our computational formula:

$$Cov(X, Y) = E[XY] - E[X]E[Y] = E[X]E[Y] - E[X]E[Y] = 0 \quad \blacksquare$$

The Critical Converse Warning

The converse is **not** generally true. If $Cov(X, Y) = 0$, the variables are uncorrelated, but they are **not necessarily independent** (since covariance

Exercise 1: Discrete Covariance Calculation

Let X and Y be discrete random variables representing the outcome of a simple system. Their joint PMF, $p(x, y)$, is defined by the following table:

	$Y = 1$	$Y = 2$
$X = 0$	0.1	0.4
$X = 1$	0.3	0.2

Your Task

1. Find the marginal PMFs of X and Y.
2. Compute the expected values $E[X]$ and $E[Y]$.
3. Compute the covariance $Cov(X, Y)$.

Exercise 2: Continuous Covariance Calculation

Let X and Y be continuous random variables. Their joint density function is defined as:

$$f(x, y) = \begin{cases} x + y & \text{if } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Your Task

1. Compute the expected values $E[X]$ and $E[Y]$.
2. Compute the expected product $E[XY]$.
3. Compute the covariance $Cov(X, Y)$.

The Cauchy-Schwarz Inequality

The relationship between the expectations of products and individual second moments is bounded by a fundamental mathematical law.

Theorem

Cauchy-Schwarz Inequality For any two random variables X and Y with finite variances:

$$(E[XY])^2 \leq E[X^2]E[Y^2]$$

In terms of deviations from their respective means, this is equivalent to:

$$(Cov(X, Y))^2 \leq Var(X)Var(Y)$$

Physical Implication

This inequality guarantees that covariance cannot grow infinitely large. It is bounded strictly by the individual spreads (variances) of the two variables, which allows us to define a standardized scale for relationships.

The Correlation Coefficient: Standardizing Covariance

While covariance tells us the direction of a linear relationship, its numerical value depends on the arbitrary units of X and Y. To fix this, we normalize it.

Definition

Correlation Coefficient The correlation coefficient of two random variables X and Y, denoted by $\rho(X, Y)$ (or simply ρ), is defined as:

$$\rho(X, Y) \triangleq \frac{Cov(X, Y)}{\sigma_X \sigma_Y} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

Core Properties of Correlation

By direct application of the Cauchy-Schwarz inequality:

- **Boundedness:** $-1 \leq \rho(X, Y) \leq 1$.
- **Perfect Linear Relationship:** $\rho = 1$ indicates a perfect positive linear trend; $\rho = -1$ indicates a perfect negative linear trend.